how great Haswell's claims are until Intel officially unveils the platform and new devices at this year's Computex event in Taipei, Taiwan, one cannot dispute the fact that Intel's hardware performance is definitely better compared to AMD and ARM, but where it lacks is power efficiency. What good are powerful products if they consume battery as quickly and won't last long enough? And the consumer so far has voted for better battery life more than better processing power, as long as mobile computing goes. That's why we are buying tablet and smartphones left, right and centre. While these ARM-based devices are now working towards increasing their performance headroom without compromising on their silver bullet – the battery life – Intel's in an opposite race to maximise its platform's power savings with Haswell's microarchitecture. And it's sparing no one, targeting everything from PCs, laptops, tablets, smartphones and hybrid devices.

Later this year, expect ultrabooks to become thinner and run cooler than the lot released last year. Also, you'll have a whole host of permutations and combinations to consider before buying a laptop or ultrabook, as Haswell not only has newer CPUs to consider but also five-levels of on-chip graphics (refer to the box alongside). Of course, it means greater and finer choice for consumers before buying an Intel product. We expect some Windows 8 ultrabooks or hybrids to reach near-tablet like battery efficiency (the iPad will still remain untouched, we think), and if that happens Windows 8 tablets could start flooding the market sooner than expected.

Moving on from Intel, we'll also see the next-generation of graphics from AMD and NVIDIA released for eager consumers – the AMD Radeon HD 8000 series, AMD Radeon HD 8000M, NVIDIA GeForce GTX 700 and NVIDIA GeForce GTX 700M graphics chips. From what we know, AMD's plan of launching the Radeon HD 8000 series of graphics cards for desktops and mobile platforms has run into unexpected delays – AMD's true Radeon HD 8000 cards won't be available until towards the end of this year, until then the company will only sell rebranded HD 7000 chips for desktop and mobile manufacturers until their true successor is revealed closer to Christmas later this year. This could be because of the custom GPU that AMD's providing on the Sony PS4 and Microsoft Xbox 720 slated for release later this year, as well. This unexpected hiccup in AMD's plans has allowed NVIDIA to steal a lead on the troubled graphics chipmaker, where it could dominate the PC market with first mover advantage. NVIDIA was supposed to have its official GeForce 700 series cards unveiling for desktop and laptop PCs last month, but probably it may happen at Computex in June. A few months ago, three GeForce GTX 700 series cards' names (and their specs) were leaked: the NVIDIA GeForce GTX 780, GTX 770 and GTX 760 Ti. It's also the chance for NVIDIA's next-gen CUDA compute architecture to shine, the Kepler GK110 – the largest GPU in terms of transistor count. According to the leak, the GTX 780 may actually be the successor to the extremely impressive Titan GPU we reviewed couple of months ago. The GTX 760 Ti is largely speculated to be almost similar to the existing GTX 670, with perhaps slightly different clock speeds to spice things up a bit. We, obviously, can't wait to test these graphics cards when they release later this year.


Ouya-controller-and-platform

## * 802.11ac: The fifth generation of Wi-Fi

The switch from wired to wireless networks was a big step as far as the advent of mobile computing goes, and we can safely say that wireless networks have truly revolutionized the way we work in the 21st century. To make life even more enjoyable, the fifth generation of Wi-Fi products have started rolling out this year, networking products that support the 802.11ac Wi-Fi protocol -- the successor to current favourite, Wi-Fi 802.11n wireless protocol. So why should you be excited about "5G Wi-Fi" or 802.11ac networking standard? We explain with the following table

| Enhancement comparison: 802.11n / 802.11ac | | |
|---|---|---|
| | IEEE 802.11n | IEEE 802.11ac |
| Frequency Band | 2.4 GHz and 5 GHz | 5 GHz only |
| Channel Widths | 20, 40 MHz | 20, 40, 80 MHz (160 MHz optional) |
| Spatial Streams | 1 to 4 | 1 to 8 total up to 4 per client |
| Multi-user MIMO | No | Yes |
| Single Stream [1x1] Max data rate | 150 Mbps | 450 Mbps |
| Three Stream [3x3] Max data rate | 450 Mbps | 1.3 Gbps |

The latest fifth generation of Wi-Fi standard not only boosts bandwidth by more than twice the capacity of the older 802.11n, it also allows more people to connect to a node simultaneously and get uninterrupted data. While 802.11ac routers will be backward compatible with 802.11n devices, interacting most of them on the 2.4 GHz frequency, all the 802.11ac compatible devices will only exchange data over the less-cluttered 5 GHz frequency band. Also, while it's great to be happy about higher speeds on Wi-Fi, it's important not to get carried away with 802.11ac just yet. Routers boasting 802.11ac compatibility have just started rolling out and most of them are extremely expensive; what's more devices (smartphone, tablets, laptops, etc) sporting 802.11ac adapters haven't launched in the market yet. But all that will change later this year. Here's to greater wireless data speeds!